

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND MEASUREMENTS

LECTURE-6

PHYSICS

WAJAHID^{Sir}



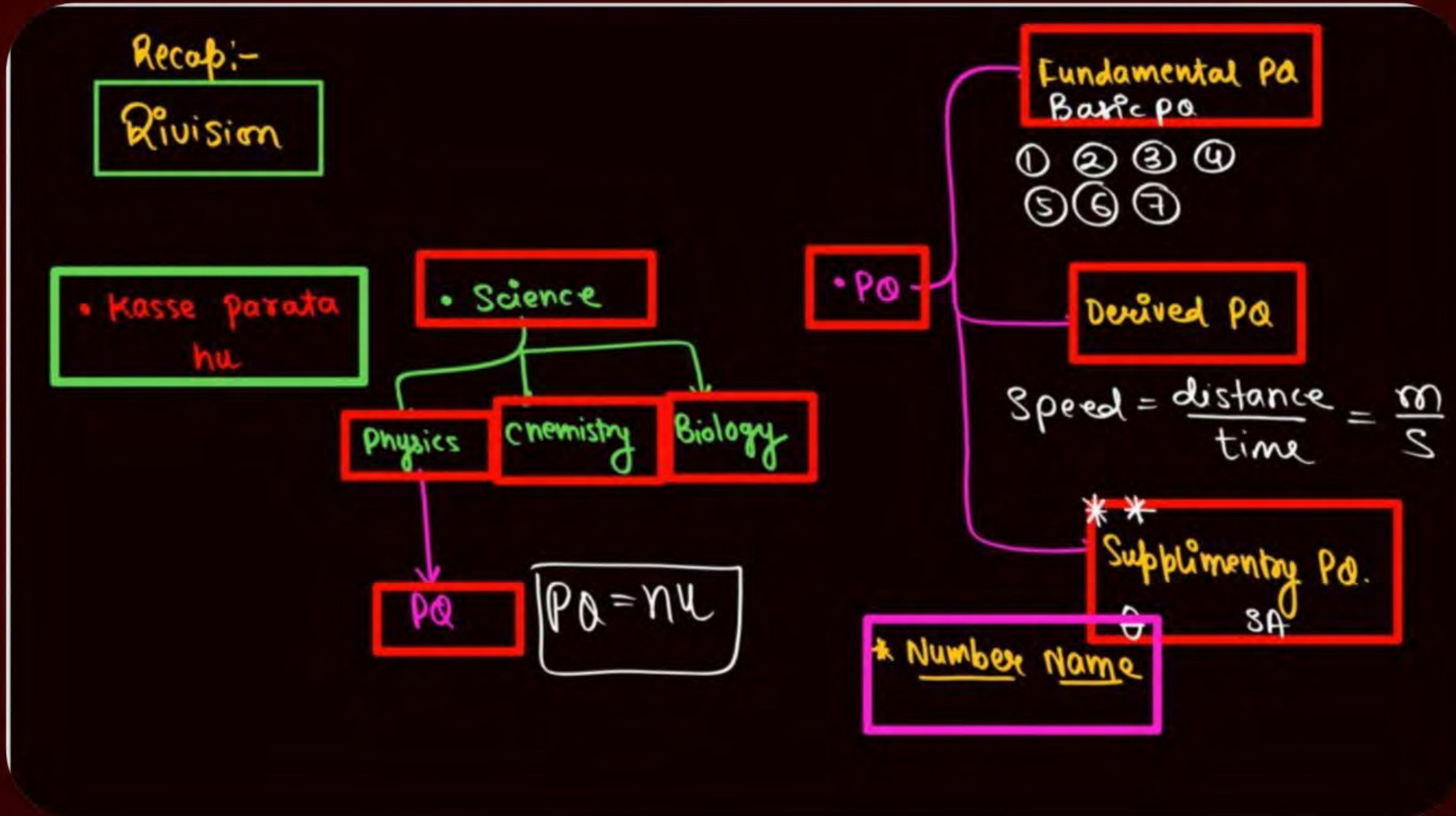
Topics To Covered

Revision

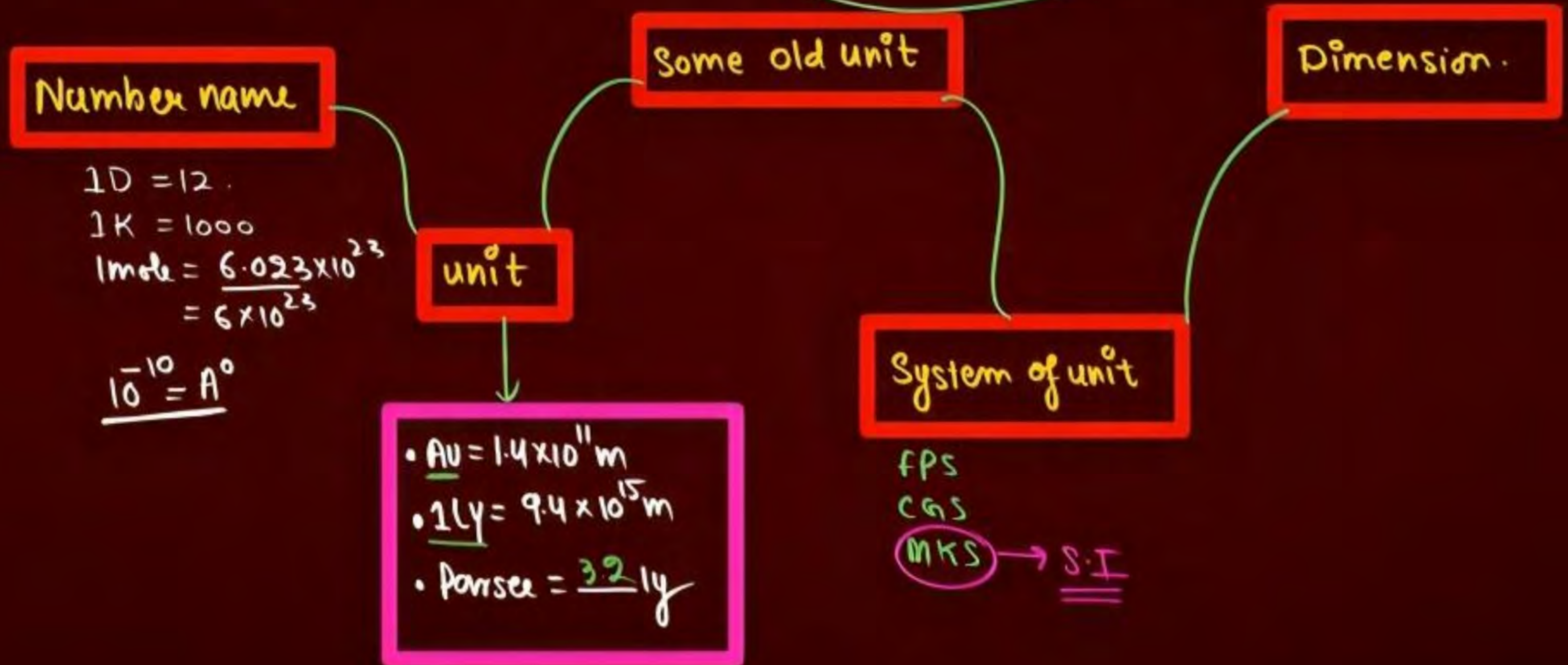
1. Conversion of one unit into another unit
2. Limitation of dimension
3. Significant Figure (story)

“Practice makes
you better every
day.”

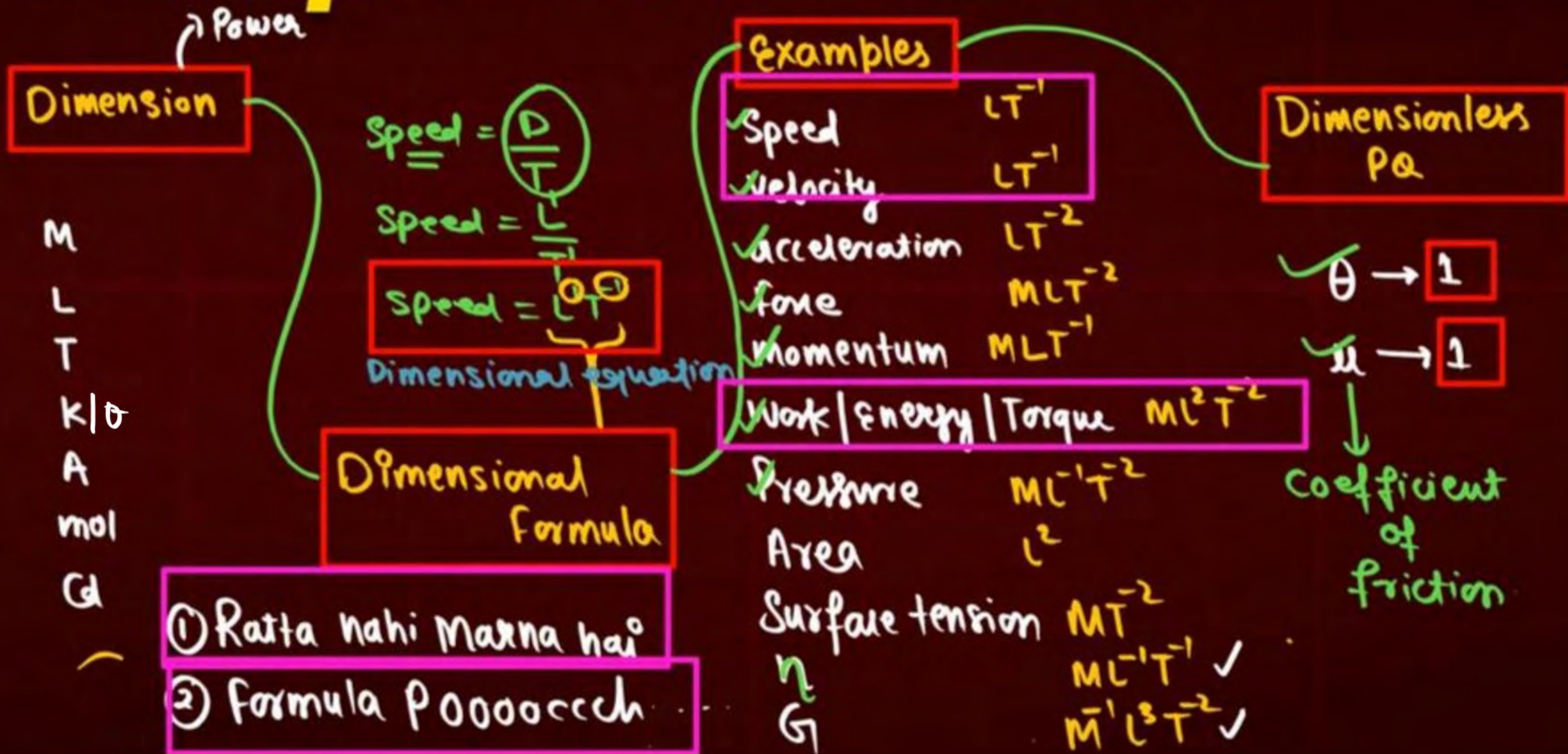
Recap From 1st Lec.



Recap From 2. Lec.



Recap From 3rd Lec.



Recap From 4th Lec.

Application
of
Dimension

a ✓

b ✓

c

d

b To check the
Correctness of
Physics Formula

Law of
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

a To find out the
Dimensional
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{M}^{-1}\text{L}^3\text{T}^{-2}$$

① $v = u + at$

$\downarrow$
 LT^{-1}

② $s = ut + \frac{1}{2}at^2$

$\downarrow$
 L

③ $v^2 = u^2 + 2as$

$\downarrow$
 L^2T^{-2}

④ $F = \frac{mv^2}{r}$

$\downarrow$
 MLT^{-2}

$\downarrow$
 MLT^{-2}

Recap From 5th Lec.



Application of
Dimension

© To derive the
physics formula

Time period of pendulum

• let

$$T \rightarrow m^a l^b g^c$$

$$M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$$

$$a=0 \quad b=\frac{1}{2} \quad c=-\frac{1}{2}$$


$$2\pi = 4\pi \sqrt{l/g}$$
$$T = 2\pi \sqrt{l/g}$$

Stoke law

• let

$$F \rightarrow \eta^a r^b v^c$$

$$MLT^{-2} \rightarrow (ML^{-1}T^{-1})^a L^b (LT^{-1})^c$$

$$a=1 \quad b=1 \quad c=1$$

$$F = 6\pi \eta r v$$

d) Conversion of one unit into another unit

Dimension Analysis used in conversion of one unit system to another unit kaise

• ek Formula banana pata hai kaise
us main n_2 value nikalni part hai

MKS

S.I System

Dimension

C.G.S

$$1\text{m} = 100\text{cm}$$

$$1\text{kg} = 1000\text{gm}$$

$$1\text{sec} = 1\text{sec}$$

$$1\text{m/s} \rightarrow 100\text{cm/s}$$

MKS

S.I System

C.G.S

$$\text{kg m/s}^2 / 1\text{N}$$

$$= 100000 \text{ dynes}$$

$$\text{kg m}^2/\text{s}^2 / 1\text{Joule} = 100000000 \text{ erg}$$

$$\text{gm cm}^2/\text{sec}^2$$

$$\text{gm cm/sec}$$



$$F = ma$$

MKS (S.I.) $\rightarrow F = \text{Kg} \frac{\text{m}}{\text{s}^2} = \text{Newton}$

C.G.S. $\rightarrow F = \text{gm} \frac{\text{cm}}{\text{s}^2} = \text{dyne}$

$$W = Fx$$

S.I (MKS) $\rightarrow W = \text{Kg} \frac{\text{m}}{\text{s}^2} \times \text{m}$
 $W = \text{Kg} \frac{\text{m}^2}{\text{s}^2} = \text{Joule}$

C.G.S. $\rightarrow W = \text{gm} \frac{\text{cm}}{\text{s}^2} \times \text{cm}$
 $W = \text{gm} \frac{\text{cm}^2}{\text{s}^2} = \text{erg}$

• $1 \text{ m}^1 = \frac{n_2=100 \text{ cm}^1}{n_2}$ (Ab Main Formula Bana Raha hu)

$$\downarrow \quad \quad \downarrow$$

$$PQ = PQ$$

$$n_1 \underline{U_1} = (n_2) \underline{U_2}$$

$$n_1 M_1^a L_1^b T_1^c = (n_2) \underbrace{M_2^a L_2^b T_2^c}_{\text{change}}$$

$$\frac{n_1 M_1^a L_1^b T_1^c}{M_2^a L_2^b T_2^c} = n_2$$

$$n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c = n_2$$

n_2

* While changing the system of unit power (Dimension) do not change

$$1 \text{ kg}^1 \frac{\text{m}^1}{\text{s}^2} = \frac{100000}{n_2} \text{ gm}^1 \frac{\text{cm}^1}{\text{sec}^2}$$

$\uparrow 10^5$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

Is formula ko kam se use karna hai

Batata hu



QUESTIONS

Convert 1 Newton into 10⁵ 100,000 n₂ dyne?

$$F \sim M^1 L^1 T^{-2}$$

$$a=1 \quad b=1 \quad c=-2$$

PYQ
2019

$$1 \text{ N} = \text{ } n_2 \text{ dyne}$$

Sol. $n_1 = 1$ $n_2 = ?$

(MKS) S.I $\rightarrow$ C.G.S

kg | m/s $\rightarrow$ gm | cm/s

$$a=1$$

$$b=1$$

$$c=-2$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

$\xrightarrow{\text{S.I (MKS)}}$
 $\xrightarrow{\text{C.G.S}}$

$$n_2 = 1 \left(\frac{1000 \text{ gm}}{1 \text{ Kg}} \right)^1 \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^1 \left(\frac{1 \text{ Sec}}{1 \text{ Sec}} \right)^{-2}$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

$$n_2 = 1 \left(\frac{1000 \text{ gm}}{\cancel{\text{gm}}} \right)^1 \left(\frac{100 \text{ cm}}{\cancel{\text{cm}}} \right)^1 \left(\frac{\cancel{1 \text{ Sec}}}{\cancel{1 \text{ Sec}}} \right)^{-2}$$

$$n_2 = 1 \times (1000)^1 \times (100)^1 \times (1)^{-2}$$

$$n_2 = 1 \times 1000 \times 100 \times 1$$

$$n_2 = 100,000 = 10^5$$

$$1^{-2} = \frac{|x|^{-2}}{1} = \frac{1}{|x|^2} = \frac{1}{1^3} = \frac{1}{1} = 1$$

QUESTIONS

$$1\text{N} = \frac{10^5}{9.8} \text{ dynes.}$$

$$1\text{N} = 1,00,000 \text{ dyne}$$

$$\underline{1\text{N}} = 10^5 \underline{\text{dyne}}$$



QUESTIONS

Convert 1 Joule = 10⁷ erg.

$$1 \text{ Joule} = \frac{n_2}{n_1} \text{ erg}$$

MKS C.G.S

kg m sec gm cm sec

$$n_1 = 1$$

$$a = 1$$

$$b = 2$$

$$c = -2$$

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

MKS C.G.S

$$\text{Energy} = M^1 L^2 T^{-2}$$

$$M^a L^b T^c$$

2021
PVQ

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$n_2 = 1 \left[\frac{1 \text{ Kg}}{1 \text{ gm}} \right]^1 \left[\frac{1 \text{ m}}{1 \text{ cm}} \right]^2 \left[\frac{1 \text{ sec}}{1 \text{ sec}} \right]^{-2}$$

$$n_2 = 1 \times \left[\frac{1000 \text{ gm}}{1 \text{ gm}} \right]^1 \left[\frac{100 \text{ cm}}{1 \text{ cm}} \right]^2 \left[\frac{1 \text{ sec}}{1 \text{ sec}} \right]^{-2}$$

$$n_2 = 1 \times (1000)^1 \times (100)^2 \times (1)^{-2}$$

$$n_2 = 1000 \times 10,000 \times 1$$

$$n_2 = 10,000,000$$

$$n_2 = 10^7$$



QUESTIONS

M-1

MKS

Convert the value of $G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$ into CGS system using dimensional analysis.

P4Q 2025

Sol $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$m^2 = 1m^2$$

$$G = 6.67 \times 10^{-11} \times 10^5 \text{ dyne} \times (100 \text{ cm})^2$$

$$G = \frac{6.67 \times 10^{-11} \times 10^5 \times 10,000 \text{ dyne cm}^2}{1000 \times 1000 \text{ gm}^2}$$

QUESTIONS

M-1

Convert the value of $G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$ into CGS system using dimensional analysis.

P4Q 2025

$$G = 6.67 \times 10^{-11} \times 10^2 \times 10^1 \text{ dynes cm}^2$$

$$G = 6.67 \times 10^{-11} \times 10^3 \text{ dynes cm}^2 \text{ gm}^2$$

$$G = 6.67 \times 10^{-11+3} \text{ dynes cm}^2 \text{ gm}^2$$

$$G = 6.67 \times 10^{-8} \text{ dynes cm}^2 \text{ gm}^2$$

$$G = 6.67 \times 10^{-8} \text{ dynes cm}^2 \text{ gm}^2$$

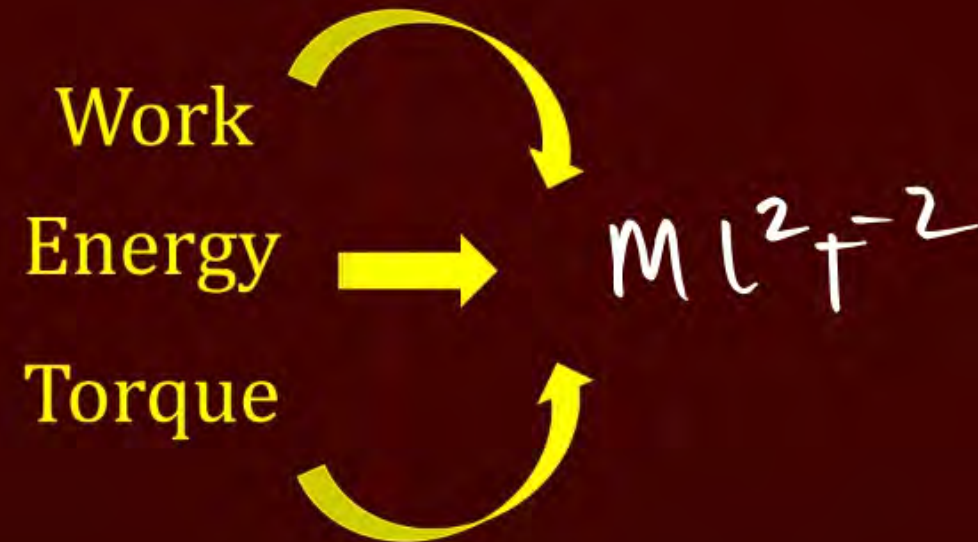
Limitation of Dimensions

(i) We cannot derive the **constt**

$$T = 2\pi \sqrt{l/g}$$

$$F_v = 6\pi\eta r v$$

(ii) Dimension are not unique



Limitation of Dimensions

(iii) We cannot derive the formula with $\checkmark/+/-$

$$V = U + at$$

$$S = Ut + \frac{1}{2} at^2$$

$$V^2 = U + 2as$$

$$V^2 - U^2 = 2as$$

$$\textcircled{1} V = U \textcircled{+} at$$

$$\textcircled{2} S = Ut \textcircled{+} \frac{1}{2} at^2$$

$$\textcircled{3} V^2 = U^2 + 2as$$

$$V^2 \textcircled{-} U^2 + 2as$$

$$T = 2\pi \times \sqrt{\frac{l}{g}}$$

$$f = 6\pi \eta \times r \times v$$



Limitation of Dimensions

(iv) We cannot derive the formula with **trigonometric function.**

$$T = \frac{2u \sin \theta}{g}$$

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

$$y = A \sin(\omega t + \phi)$$

$\sin \theta$
 $\cos \theta$

$\tan \theta$

$\operatorname{cosec} \theta$

$\sec \theta$

$\cot \theta$



QUESTIONS

(Revision)

MCA

M-1

PYQ 2025

Plank's constant has same dimensions as that of:

- 1 Energy ✓
- 2 Momentum ✓
- 3 Frequency ✓
- 4 Angular momentum

$$h = \epsilon T$$

$$h = ML^2 T^{-2} T^{-1}$$

$$h = ML^2 T^{-1}$$

QUESTIONS

Stress has the dimensions same as that of:

PYQ 2024

$$\text{Stress} = \frac{F}{A}$$

$$\text{Pressure} = \frac{F}{A}$$

- 1 Energy
- 2 ☒ Pressure
- 3 Torque
- 4 Power

Young modulus of elasticity $\longrightarrow$ Stress

Limitation of D

$$1J = 10^7 \text{ erg}$$

Lecture no. 6

$$n_2 = n_1 \left[\frac{m_1}{m_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$\Delta N = 10^5 \text{ dy}$$

HOMEWORK

Convert **10 Newton** into **dyne** using dimensional analysis.

QUESTIONS

Convert 5 Newton into dyne using dimensional analysis.

Thank
You